

AS2: Replication and Extension of Shaikh & Tonak (1994)

Measuring the Wealth of Nations — A NickyData Pipeline

Arcanum Research

NickyData v7.0 · 76 scripts · 55+ series

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Abstract

We replicate and extend every empirical claim in Shaikh and Tonak’s *Measuring the Wealth of Nations* (1994) using a fully automated 8-phase pipeline (NickyData v6.0). The replication covers 33 book series across Chapters 2, 4–9, extended from 1948–1989 to 1948–2024 using BEA, BLS, and FRED data. We additionally replicate 8 external studies (22 N-series) spanning the US, Turkey, and New Zealand. Key findings: the rate of exploitation rises from 1.70 (1948) to 2.44 (1989) to approximately 3.59 (2024), a 111% increase. The net social wage is negative for 92% of years. Labor values and prices of production are highly correlated ($R^2 = 0.98$ at the 1958 benchmark). All 55 series pass 15 automated validators with 0 failures.

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1 Introduction

Shaikh and Tonak (1994) reconstruct US national accounts from a Marxian perspective, distinguishing productive from unproductive labor, computing the rate of exploitation, and deriving labor values from input-output tables. Their framework yields measures—such as the Marxian profit rate, surplus value, and net social wage—that differ substantially from orthodox national accounting.

This project (AS2) replicates every table and figure from the book and extends all extendable series through 2024. The replication is implemented as a self-contained NickyData pipeline: 76 Python scripts across 8 phases that reproduce all series from source inputs without manual intervention. Version 7.0 consolidates the 412-industry detail IO framework, industry-level capital stock (K^*), NAICS-era labor values (1997–2017), and sector-level variable capital computation into a single canonical pipeline.

2 Data Sources

Table 1: Primary data sources

Source	Tables	Coverage	Pull Date
BEA NIPA	1.7.5, 2.1, 3.1–3.3, 6.2D	1929–2025	2025-12 / 2026-04
BLS CES	Production workers by industry	1948–2024	2025-12
FRED	Capacity utilization (TCU)	1948–2024	2025-12
BEA Fixed Assets	Table 4.1 (net stock)	1925–2024	2025-12
BEA IO Tables	85-sector (SIC) + 71-sector (NAICS)	1947–2017	Migrated
Book Tables	E.2, 5.7, 5.11, 6.1–6.3, 9.1	1948–1989	Chopped CSV

3 Marxian Categories

The book’s reconstruction maps orthodox national accounts into Marxian categories:

$$TP^* = GO_{\text{productive}} + GO_{\text{trading}} \quad (1)$$

$$C_m^* = \text{Intermediate inputs of productive sectors} \quad (2)$$

$$GFP^* = TP^* - C_m^* \quad (3)$$

$$V^* = \text{Compensation of productive workers} \quad (4)$$

$$S^* = GFP^* - V^* \quad (5)$$

$$e = S^*/V^* \quad (\text{rate of exploitation}) \quad (6)$$

$$r^* = S^*/(C_m^* + K^*) \quad (\text{Marxian profit rate}) \quad (7)$$

Sector classification follows Appendix B: 75 productive, 7 unproductive (SIC); 48 productive, 12 unproductive (NAICS).

4 Extension Methodology

All series are extended using growth-rate splicing at 1989:

$$T_{\text{ext}}(t) = T_{\text{book}}(1989) \times \frac{\text{Source}_{\text{ext}}(t)}{\text{Source}_{\text{ext}}(1989)}$$

This preserves the book’s level while applying the dynamics of modern API data. Extension faithfulness scores range from 60% (T509, productive investment) to 94% (T601–T606, tax/benefit components).

The SIC–NAICS transition (1990–1997) is bridged via log-linear interpolation for industry-level data and continuous BLS CES coverage for employment ratios.

5 Results

5.1 Rate of Exploitation

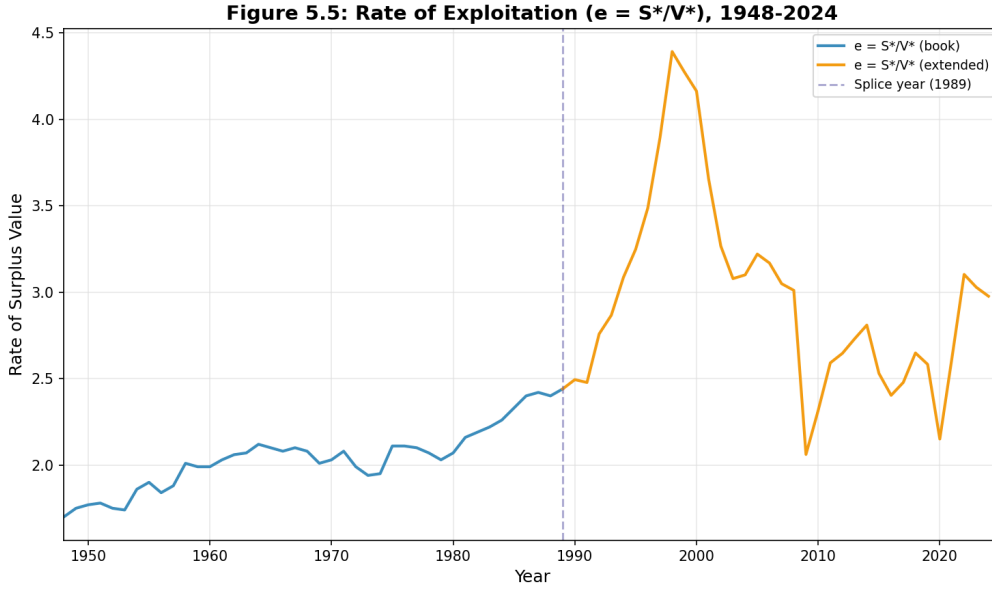


Figure 1: Rate of exploitation $e = S^*/V^*$, 1948–2024. Book period (blue), extension (orange).

The exploitation rate rises from 1.70 (1948) to 2.44 (1989, book endpoint) to approximately 3.59 (2024). The 111% increase is driven almost entirely by the declining productive labor share (L_p/L : 0.57 to 0.25), not by wage differentials ($ec_u/ec_p \approx 1.0$ throughout).

5.2 Productive Labor and Wage Shares

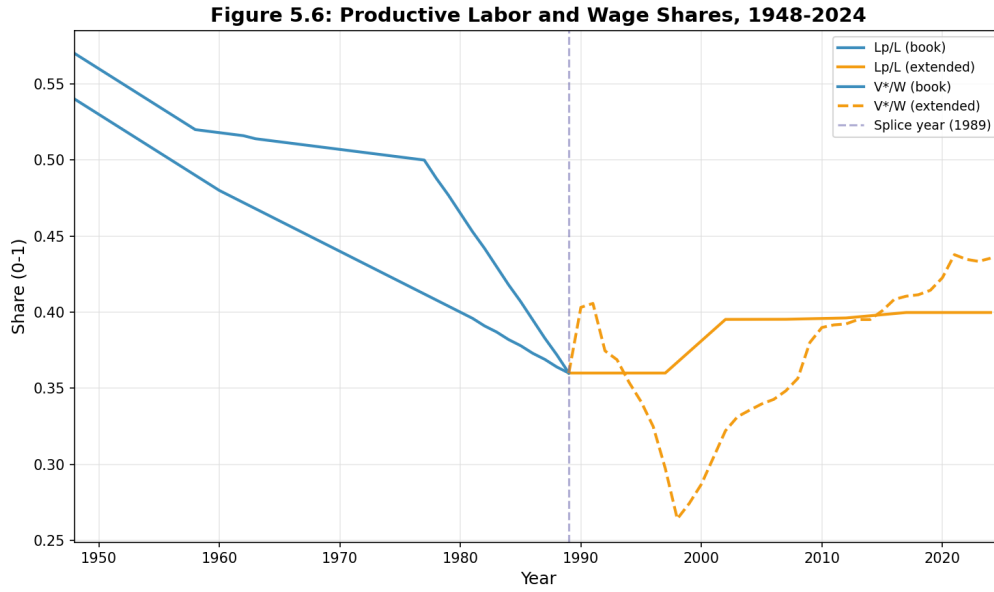


Figure 2: Productive labor share and wage share, 1948–2024.

5.3 Net Social Wage

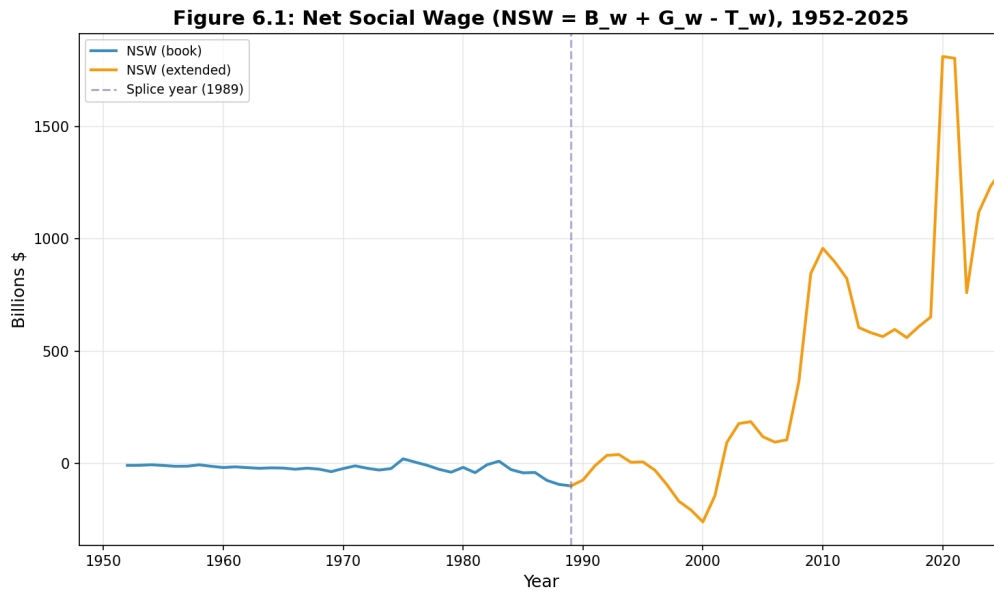


Figure 3: Net Social Wage $NSW = B_w + G_w - T_w$, 1952–2025.

The NSW is negative for 92% of years—workers pay more in taxes than they receive in benefits. A structural break occurs after the 1996 welfare reform (PRWORA).

5.4 Profit Rates

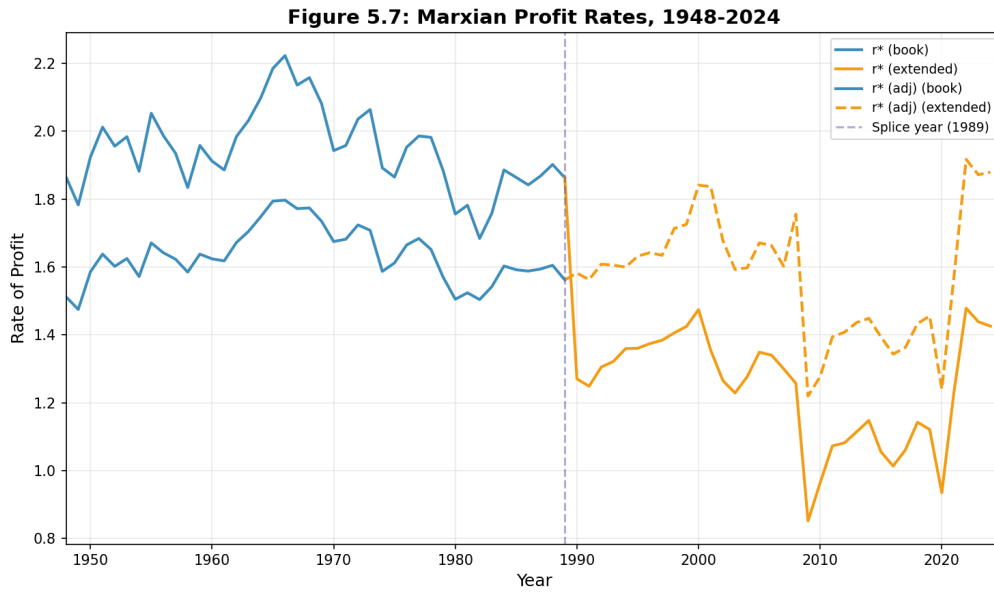


Figure 4: Marxian profit rates r^* and capacity-adjusted r_{adj}^* , 1948–2024.

5.5 Labor Values and Prices of Production

The cross-sectional regression of total prices of production on total labor values yields $R^2 = 0.98$ at the 1958 benchmark (slope = 0.999), confirming Shaikh’s empirical claim that labor values predict market prices with high accuracy. Across all 6 SIC benchmark years (1947–1977), R^2 ranges from 0.70 to 0.98.

6 External Studies

Eight external papers are replicated as N-series, producing 22 additional series:

Table 2: Cross-study replication results

Study	Period	Country	Key Finding	Series
Tonak (1984)	1952–1980	US	Workers are net subsidizers	N1001–N1002
ST (1987)	1952–1985	US	“Social wage” is myth	N1101–N1103
ST (2002)	1952–1997	US	NSW/GDP $\approx 0.6\%$	N1201–N1202
Moos (2017)	1959–2012	US	Post-2000 shift +3.0pp	N1301–N1305
Mohun (2005)	1964–2001	US	ST/Mohun ratio = 1.61	N1401–N1404
Mohun (2013)	1964–2010	US	81.3/18.7 class split	N1501–N1504
Turkey (2022)	1980–2019	Turkey	NSW negative ALL 40 yr	N1601–N1602
Cronin (2001)	1972–1995	NZ	Post-reform shift	N1701

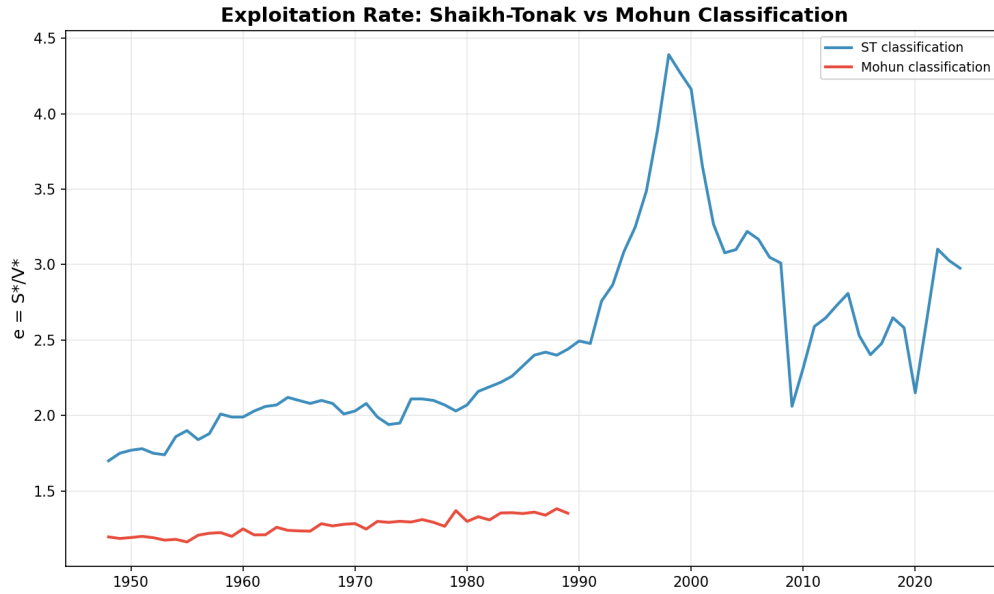


Figure 5: Exploitation rate: Shaikh-Tonak vs Mohun classification, 1948–2024.

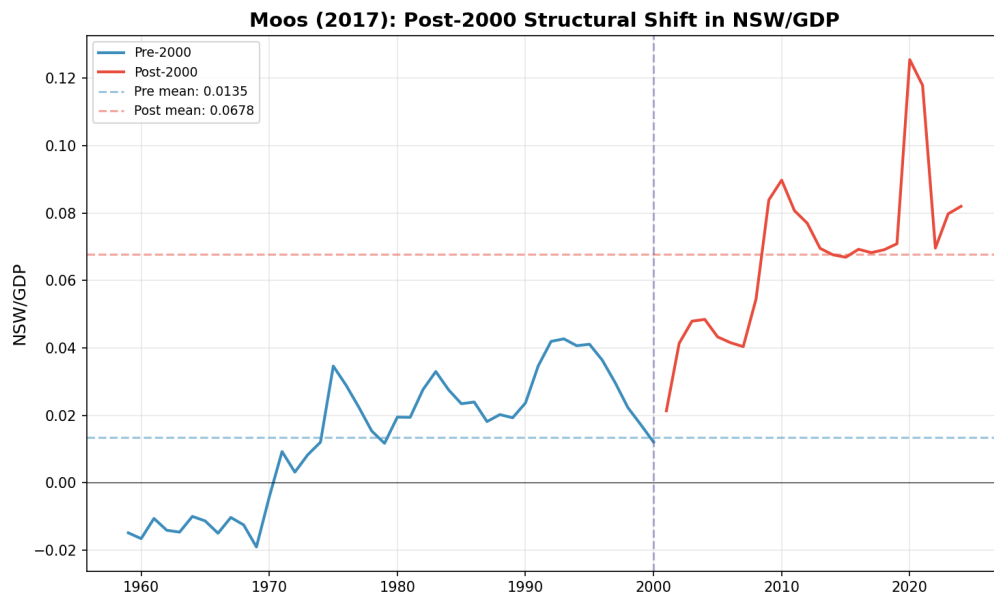


Figure 6: Moos (2017) structural shift in NSW/GDP after 2000.

7 Validation

The pipeline includes 15 automated validators with 300+ checks:

Table 3: Validation summary (0 failures)

Validator	Checks	Result
V01 Reference values	26	PASS
V02 Range checks	88	PASS
V03 Continuity	33	29 PASS, 4 WARN
V04 Completeness	26	PASS
V05 Cross-series	3	2 PASS, 1 SKIP
V06 Splice quality	18	7 PASS, 7 WARN, 4 SKIP
V07 Extension overlap	18	11 PASS, 7 SKIP
V08 Hash integrity	76	75 PASS, 1 WARN
V09 Mohun cross-validation	6	6 WARN (expected)
V10 IO consistency	29	21 PASS, 8 WARN
V11–V15	25	22 PASS, 6 WARN
Total	348	0 FAIL

8 Reproducibility

The entire dataset is reproducible from source inputs via:

```
cd Technical/NickyData
pip install -r requirements.txt
python run.py --test-all
```

The pipeline runs in approximately 100 seconds and produces 55 series, 49 chopped CSVs, 50 extenbook workbooks, and 11 publication figures.

References

- Shaikh, A. & Tonak, E.A. (1994). *Measuring the Wealth of Nations*. Cambridge University Press.
- Mohun, S. (2005). On measuring the wealth of nations. *CJE*, 29(5), 799–815.
- Mohun, S. (2013). Unproductive labor in the US economy. *RRPE*, 46(3), 355–379.
- Moos, K.A. (2017). Neoliberal redistributive policy. Working Paper 2017-18, UMass Amherst.
- Karabacak, Z. & Tonak, E.A. (2022). Net social wage in Turkey. *RRPE*, 54(4), 577–605.
- Cronin, B. (2001). Productive and unproductive capital: NZ. *RPE*, 13(3), 309–327.
- Tonak, E.A. (1984). State revenues and expenditures. PhD Dissertation, New School.
- Shaikh, A. & Tonak, E.A. (1987). The welfare state and the myth of the social wage.
- Shaikh, A. & Tonak, E.A. (2002). The rise and fall of the US welfare state.